

An insight into machine learning algorithms' accuracy in predicting mud loss rate: Data processing and screening

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Abstract

In drilled wells, a phenomenon known as "lost circulation" (LC) has been connected to problems such as "pipe sticking" and "formation damage." These worries act as a catalyst, which results in an increase in both the financial costs and the total duration of the project. The purpose of this study is to improve the process of data cleaning and screening in order to achieve the goal of increasing the accuracy of LC rate predictions made during drilling. The applied machine learning method is a component of an algorithm for supervised learning that also incorporates artificial neural networks (ANN) and support vector machines (SVM). RPM, lithology, gel10min/gel10s, Fann 600/Fann 300, pump pressure, hole size, fracture pressure, pore pressure, and northing and easting from the Marun field are some of the data that were used. Before being used to train and test machine learning models, these data sets underwent preliminary inspection and processing. Data processing and screening frequently make use of statistics such as the interquartile range and Pearson's correlation coefficient. The validation of the data was done by the utilisation of a 10-fold cross-validation procedure. The correctness and effectiveness of the models were checked to ensure that field deployment is possible without the introduction of potentially catastrophic mistakes. The RMSE and R^2 values that were calculated for each model were contrasted with one another. When it came to SVM models, the Fine Gaussian SVM model had the highest R^2 value, with 0.83 for training and 0.91 for testing; however, the Wide Neural Network model had the highest R^2 value when it came to ANN models, with 0.95 for training and 0.99 for testing. It is possible to conclude, on the basis of the findings of the study, that Wide NN is the best model for predicting LC rates in oil and gas wells. The accuracy of the model is enhanced by the cleaning and filtering of the data.

Keywords: Loss circulation; Machine learning; Mud loss rate; Support Vector Machine; Artificial Neural Network; Drilling

1.0 Introduction

Mud loss circulation (LC) cannot be avoided entirely during drilling for the extraction of oil and gas. It can cause pipe sticking, formation damage, and uncontrolled flow of wellbore fluids into the formation (Behnoud far & Hosseini, 2017; Magzoub et al., 2020). LC usually happens in wells that have high permeability and/or fractures and can also be present when there are induced fractures. Additionally, LC

has an increased possibility when the well is drilled at deeper depths or at reservoirs that are depleted with low pore pressure (Alsabagh et al., 2015).

There are four categories of LC: seepage mud losses, partial mud losses, severe mud losses, and total losses (Elkakatny et al., 2020; Jeennakorn et al., 2017; Sauki et al., 2021). The total mud losses indicate that 100% of the mud has no returns. According to Sabah et al., (2020), the three main LC rates are seepage losses that have rates of 1-10 bbl/hr, partial losses that have rates of 10-500 bbl/hr, and complete losses with rates above 500 bbl/hr. Predicting the LC rate is critical since a loss of mud circulation can result in non-productive time during drilling and lead to economic concerns, which are more prevalent when drilling activities take place in deep wells and HPHT wells.

Much research has been conducted to date to forecast these LC rates, with the majority of them employing machine learning (ML) to do so (Al-Hameedi et al., 2018; Alsaihati et al., 2022; A. Moazzeni et al., 2012; A. R. Moazzeni et al., 2010; Mohammad Ali, 2015). This includes a study conducted by Abbas et al. (2019) that showed how machine learning can be used to finalize intelligent decisions to stop or mitigate lost circulation. In the past, conventional methods to predict mud loss rates include conducting leak-off tests (LOT) and formation integrity tests (FIT). There are a few negative effects that can occur during LOT which include wellbore weakening due to hydraulically conductive cracks and leak-off tests (Wang et al., 2011). Other than that, the formation integrity test is somehow limited due to the low accuracy of the test depending on the complex well integrity and marginal errors can also frequently occur in the predictions. The common ML algorithms used for LC rate prediction are SVM and ANN and they were used extensively for enhancing the accuracy of LC rate prediction at various input parameters as shown in Table 1 (Abbas et al., 2019; Sabah et al., 2020; Sabah et al., 2019; Geng et al., 2019; Toreifi et al., 2014; Behnoud far & Hosseini, 2017).

60

Table 1 Comparison summary of the ML-based prediction methods for LC incidents.

Author	Parameter	ML model	Performance
(Geng et al., 2019)	Seismic data	Supervised ML with majority voting algorithm	Cross validation score is 0.8 (model performance).
(Abbas et al., 2019)	MW, ROP, CP, Flow rate, Lithology, inclination, solids content, RPM, WOB, YP, PV, marsh funnel viscosity, Gel 10 min, Gel 10 s, azimuth, MD and hole size.	ANN SVM	<u>ANN R^2 value</u> Accuracy – 0.87 Sensitivity – 0.89 Specificity – 0.85 <u>SVM R^2 value</u> Accuracy – 0.92 Sensitivity – 0.93 Specificity – 0.91
(Sabah et al., 2019)	Pore pressure, formation fracture pressure, mud pressure during the drilling,	MLP-NN RBF-NN	<u>R^2 Value</u> <u>MLP-NN</u> Test – 0.9073

	nit RPM, drilling time, formation type, gel strength, pump pressure, pump rate, WOB, hole diameter, hole depth, geographic cartesian coordinates, viscosity, θ_{600} , and θ_{300} .	GA-MLP ANFIS DT	Train – 0.9213 <u>RBF-NN</u> Test – 0.8445 Train – 0.85 <u>GA-MLP</u> Test – 0.826 Train – 0.8307 <u>ANFIS</u> Test – 0.8855 Train – 0.9018 <u>DT</u> Test – 0.93 Train – 0.9725
(Toreifi et al., 2014)	East, North, press depth, ROP, formation type, V_{ANN} , formation top, pump output, pump pressure, mud pressure, mud filtrate viscosity, ret solid, PV, YP, initial gel, 10 min gel and lost	MNN MLP	<u>R^2 value</u> <u>MNN</u> Model 1 – 0.891 Model 2 – 0.964 <u>MLP</u> Model 1 – 0.627 Model 2 – 0.745
(Behnoud far & Hosseini, 2017)	MW, PP, Depth and pump flow rate	ANN	<u>R^2 value</u> 0.996, 0.9991, 0.9988
(Sabah et al., 2020)	Lithology, RPM, Gel 10min/Gel 10s, pump pressure, fracture pressure, pore pressure, northing, easting, fan 600/fan 300, hole size.	MLP-COA MLP-PSO MLP-GA LSSVM LSSVM-COA LSSVM-PSO LSVM-GA	<u>R^2 value</u> 0.9381, 0.9517 0.9353, 0.9489 0.9304, 0.9442 0.9195, 0.9381 0.9424, 0.9556 0.9391, 0.9509 0.9315, 0.9451

ANN and SVM is a supervised ML algorithm that indicates that the data sets are labelled and each of the data points has a set label or target (Nabi, 2018). SVM models would usually include statistical learning theory to be used for classification and regression tasks. This is done by using the optimal separation rule where the test data is allocated into a high-dimensional feature space and employing a non-linear mapping method (Abbas et al., 2019). In ANN, the artificial neurons that act as the processing elements consist of four basic components. These components include input data, connection strengths (weights), a transfer function (activation function), and output values (Abbas et al., 2019). Each of these layers has a pre-set number of neurons. These parameters need to be tuned by trial-and-error technique. The input data to each neuron located in the hidden layers will be multiplied by the connection strengths ($w_{lj} \dots w_{nj}$), combined, and go through the transfer function that will produce the output values of the neurons. The most common ANN model used includes the feed-forward neural network, and multi-layer perceptron (MLP) (Kazemi et al., 2016).

75 Nevertheless, the accuracy of each ML algorithm is much dependent on its input data. According to Ren
 76 et al., (2022), the logging data that will be used as input for the lithology recognition model needs to be
 77 processed and is unable to be fed directly. If the data is fed directly, the degree of influence would create
 78 an error in the results. This is mainly because raw logging data provided from different logging methods
 79 would acquire different dimensions and magnitudes of attribute values. Additionally, processing the data
 80 would also help in improving the performance of machine learning methodologies and increase the speed
 81 of the learning process. To avoid systematic errors that could happen, the data would need to undergo
 82 standardization. The method for standardization that is commonly used in the past pieces of literature
 83 includes the min-max standardization method (Khodnenko et al., 2019), priori probability and quantile
 84 method (Akouemo et al., 2016 ; Vallis et.al., 2014), Akima model method (Khodnenko et al., 2019) and
 85 *fscaret* package (Szlek, 2018). This study intends to improve the accuracy of LC rate prediction by
 86 suggesting an alternative standardisation approach or data cleaning method, as well as data screening as an
 87 input to ML algorithms, in comparison to a study using the same Marun field data by Sabah et al., (2020).
 88 The novelty of this research is demonstrated by the data processing and screening procedures, which differ
 89 from previous studies. ML techniques such as ANN and SVM models were examined to establish the best
 90 model for forecasting LC rate.

91 **2.0 Methodology**

92 *2.1 Data Acquisition*

93 The drilling dataset used in this investigation was derived from the work of Sabah et al. (2020). The
 94 dataset was obtained as a supplemental file from Mendeley Data. The dataset that is now available is as
 95 follows:

- 96 • Northing
- 97 • Easting
- 98 • Depth (m)
- 99 • Meterage
- 100 • Drilling time (hr)
- 101 • Formation type
- 102 • Hole size
- 103 • Weight on bit (1000lb)
- 104 • Flow rate (gpm)
- 105 • Plastic viscosity
- 106 • Retort solid (%)
- 107 • Mud weight (pcf)

- Pore pressure (psi)
- Fracture pressure (psi)
- Fann 600/Fann 300
- Gel 10 min/Gel 10s
- Pump pressure (psi)
- Rotations per minute
- Mud loss (bbl/hr)

These datasets were compiled from daily mud reports (DMRs), final well reports (FWRs), and daily drilling reports (DDR) from the 305 wells drilled in the Marun oil field in the past. There are a total of 2820 data sets used in the predictive model construction. The Marun oil field is located in Iran, to the south of the northern Dezful embayment, at 30°51'13" N and 49°50'19" E. Carbonates and shale make up the Asmari and Bangestan oil reserves. The distinction between the two oil reserves is that Asmari has sandstones, whereas Bangestan does not (Sabah et al., 2020). Because mud loss is more common in carbonate reservoirs, the Marun oilfield dataset was used for the study.

2.2 Data Processing

The acquired data had been processed to remove any outliers. Outliers are data values that are significantly higher or lower than the average (Deangelis & Ayers, 2009). To reduce noise in the data, these outliers must be deleted. When there is excessive noise in the dataset, the accuracy of the models suffers. Outliers were identified using Python programming, and boxplots of the parameters were made to determine the number of outliers in each parameter. Python programming was used to remove outliers in each parameter, and the processed dataset was used in data screening. The outliers were removed by calculating each parameter's interquartile range (IQR). The following equation can be used to compute the interquartile range:

$$\begin{cases} Z_u - Q_3 + 1.5(Q_3 - Q_1) \\ Z_d - Q_1 + 1.5(Q_3 - Q_1) \end{cases} \quad (1)$$

where: Q_1 refers to the lower quartile (25% quantile of data), Q_3 refers to the upper quartile (75% quantile of data), Z_u is the upper quartile with 1.5 times the value of IQR and Z_d is the lower quartile minus 1.5 times the value of IQR (Ren et al., 2022). IQR was calculated to find the difference between the upper and lower quartile range. As a standard, 1.5 times IQR is used and the points higher than Z_u and lower than Z_d is classified as an outlier.

140 2.3 Data Screening

141 In addition to noise reduction, redundancy must be reduced to improve model accuracy. The parameters
 142 are ranked by their significance in the data screening. The parameters can be chosen using filter methods
 143 and correlation coefficients. Pearson's correlation coefficient was employed in this study. Pearson's
 144 correlation coefficient can measure the linear association strength between two parameters denoted by r
 145 (Benesty et al., 2008). A line of best fit was drawn, and the r value would indicate the distance between the
 146 lines. The r value would range between +1 and -1. The strength scale of the coefficient is shown in Table
 147 2 below.

148 **Table 2:** Pearson's Correlation Coefficient: Strength Indicator (Sauki et al., 2020)

Strength of Association	Coefficient, r	
	Positive	Negative
Very high correlation	0.90 – 1.00	-0.9 – -1.0
High correlation	0.7 – 0.9	-0.7 – -0.9
Moderate correlation	0.5 – 0.7	-0.5 – -0.7
Low correlation	0.3 – 0.5	-0.3 – -0.5
Negligible correlation	0.0 – 0.3	-0.0 – -0.3

149
 150 Microsoft Excel was used for the correlation. The parameters with moderate to high positive/negative
 151 correlation were chosen to be employed in the prediction models.

152 2.4 ML Algorithm

153 2.4.1 SVM

154 SVM was built with the Regression Learner app in MATLAB to pinpoint the most suitable SVM models
 155 for predicting mud loss rates. The SVM models includes a variety of kernel functions such as linear,
 156 quadratic, cubic, fine Gaussian, medium Gaussian, coarse Gaussian, and so on (Faricha et al., 2020). The
 157 following are the comparable SVM kernel functions used in the model:

- 158 • Linear SVM
- 159 • Quadratic SVM
- 160 • Cubic SVM
- 161 • Fine Gaussian SVM
- 162 • Medium Gaussian SVM
- 163 • Coarse Gaussian SVM

164 2.4.2 ANN

165 All the ANN model types were chosen to be constructed with the Regression Learner app in MATLAB
 166 to compare the best model for predicting mud loss rates. There are five (5) different types of ANN models

that were chosen for this study (Khan et al., 2022).

- Narrow NN
- Medium NN
- Wide NN
- Bilayered NN
- Trilayered NN

2.5 Developing ML Model

2.5.1 Model Build

The screened and processed dataset parameters were input as a variable in MATLAB and divided into inputs and outputs. The regression learner app in MATLAB was used to build the SVM and ANN models. The input was divided into 90:10 where 90% was used for training and 10% was used for testing. In the regression learner app, all the SVM and ANN model types were chosen for training and testing to compare the performance of each model.

2.5.2 Data validation

For data validation, 10-fold cross-validation was used. This method divides the dataset into 10 parts and uses 9 out of the 10 parts for the training subset. The left-out part was used as a testing subset. The cross-validation method was repeated 10 times with non-overlapping segments for the training and testing subsets. This feature selection step allows the optimum combination of features to be selected to allow the yield of the minimum RMSE value of the model (Sabah et al., 2020).

2.6 Model Accuracy Evaluation

The model performance was evaluated by various statistical performance indices that include RMSE, and R^2 that were used by Basarir et al. (2014) and Sabah et al. (2019).

$$RMSE = \left(\frac{1}{p} \sum_{r=1}^p (-z_r)^2 \right)^{\frac{1}{2}} \quad (2)$$

$$R^2 = \left(1 - \frac{\sum_{r=1}^p (y_r - z_r)^2}{\sum_{r=1}^p (y_r - y_{r,mean})^2} \right) \quad (3)$$

where y is actual output, z is model output and p is the amount of data in the data set. The accuracy of the proposed model would depend on the small value of RMSE and the high value of R^2 . Each step in the methodology is visualized in Fig 1 from the data acquisition and processing step to the model evaluation for field application.

1. Data Acquisition and Processing

Data Acquisition - Marun oilfield

- Northing
- Easting
- Depth
- Meterage
- Drilling time
- Formation type
- Hole size
- WOB
- MW
- Plastic Viscosity
- Retort Solid
- Pore pressure
- Fracture Pressure
- Fan 600/Fan 300
- Gel 10min/Gel 10s
- Pump Pressure
- RPM
- Mud Loss

Data Processing

- Interquartile range (remove outliers).
- Pearson's correlation coefficient (data parameter ranking).

Data Screening

- RPM
- Lithology
- Gel 10 min/Gel 10s
- Pump Pressure
- Fracture Pressure
- Pore Pressure
- Northing
- Easting
- Fan 600/Fan 300
- Formation type
- Hole Size

2. Machine Learning Algorithm

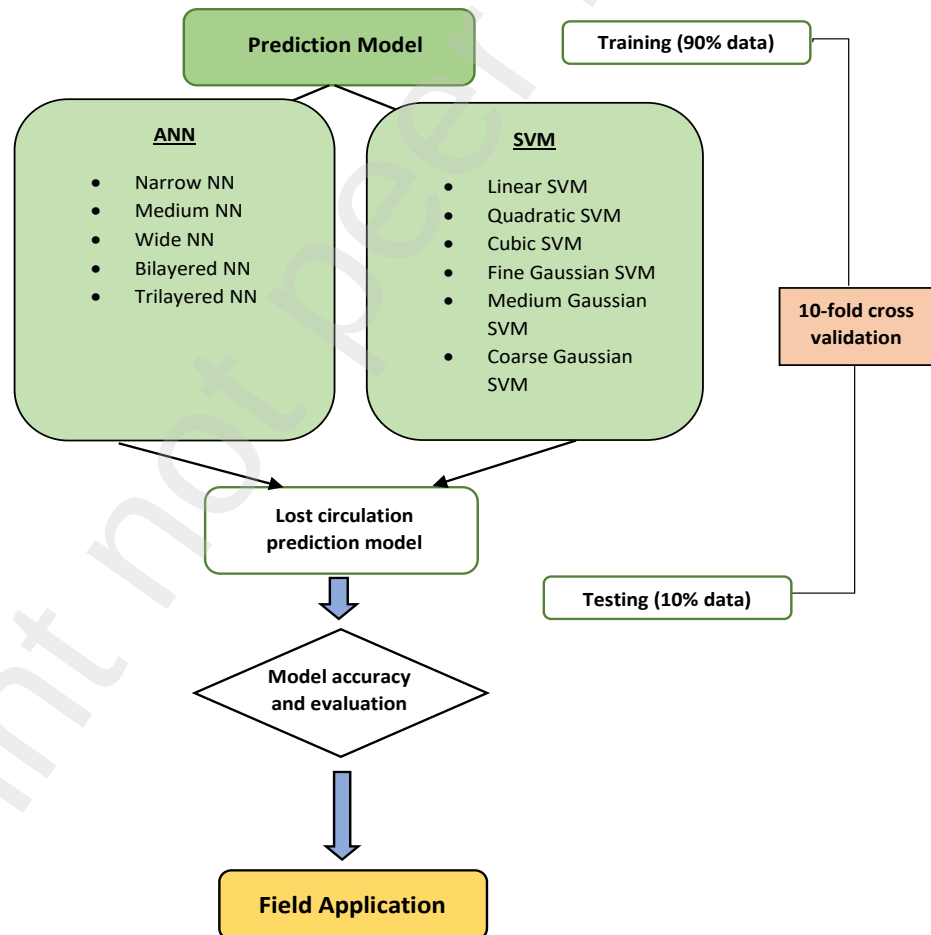


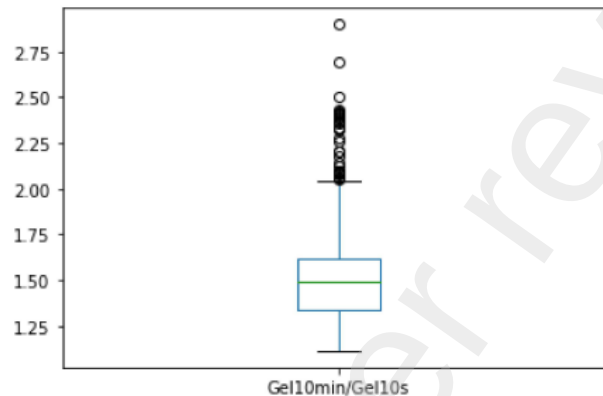
Fig 1: Flowchart for constructing prediction model.

198 **3.0 Results and discussion**

199 **3.1 Data Processing and Screening**

200 Boxplots were constructed in Python to visualize the number of outliers detected in each parameter.
201 Based on the boxplot (Fig.2), the outliers were seen to be presented in Gel 10min/Gel 10s. The data with
202 outliers were removed and a new clean dataset was produced. This was proven when the original data
203 containing 2820 rows were reduced to 2572 rows of data. The total number of outliers removed was 250.

204



205

206 **Fig 2:** Boxplot showing outliers in Gel 10min/Gel 10s

207

208 The data was then transferred as comma-separated values files and opened in Microsoft Excel. To
209 determine the ranking of the parameters, Pearson's correlation coefficient was calculated and a heatmap
210 was created. The parameters were chosen based on their degree of correlation with mud loss rate, as seen
211 in the heatmap in Fig. 3. The moderate to strong coefficient values, r (0.5, -0.5) were chosen to allow the
212 prediction model to perform better with less redundancy and noise. Fig. 4 depicts a graph of coefficient
213 values for each parameter. The graph aids in seeing the parameters that have a moderate to high link with
214 the rate of mud loss. Parameters that fall between the red lines (0.5, -0.5) were deemed to be highly
215 significant to the prediction model. Table 3 displays the most strongly linked parameters that were chosen.

216

217

218

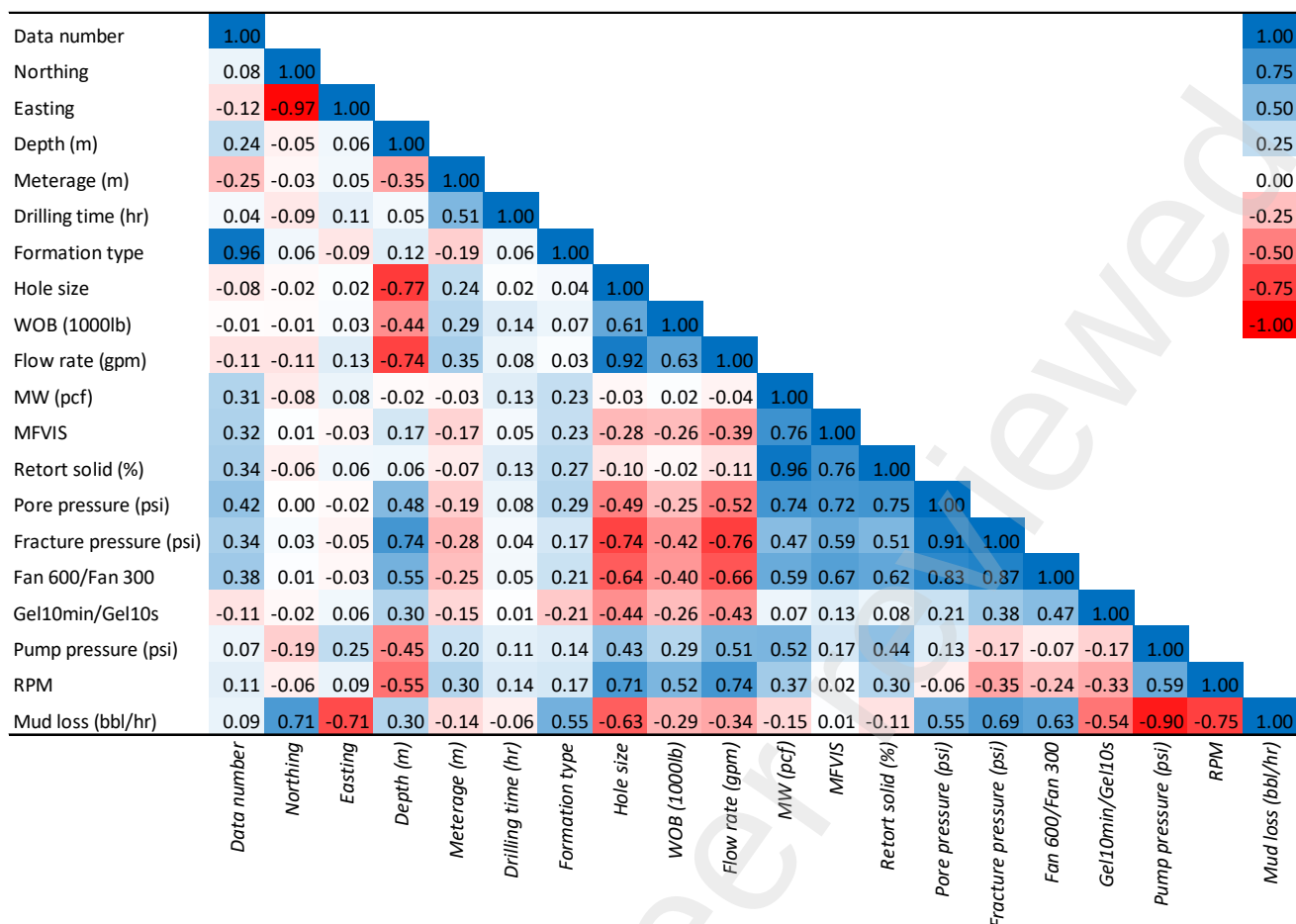


Fig 3: Heatmap of Pearson's correlation coefficient

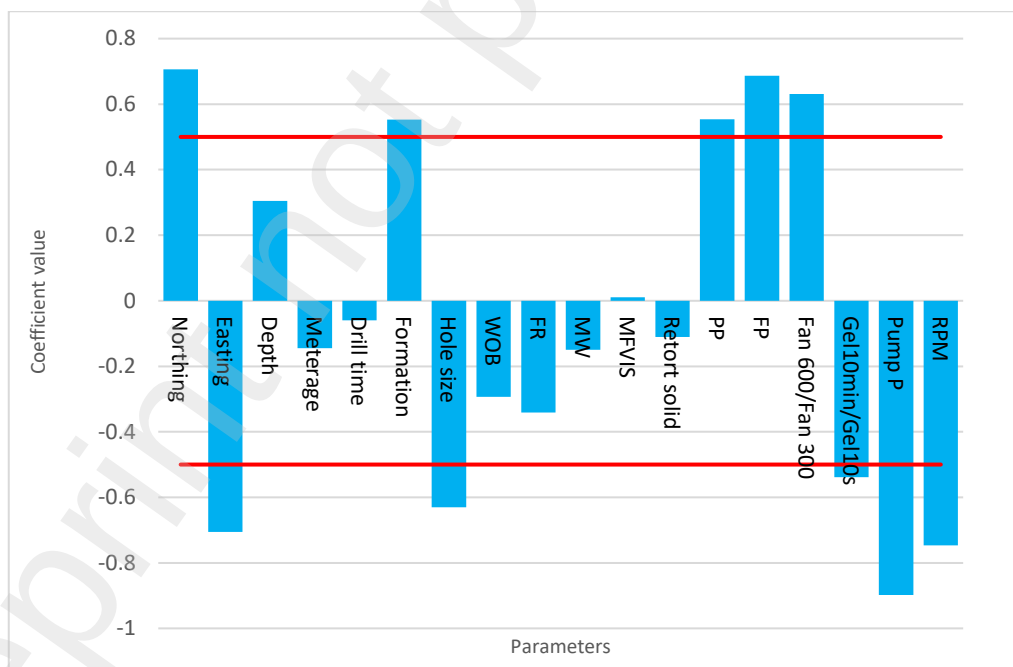


Fig. 4: Pearson's correlation coefficient

Table 3: Selected data parameters used in the predictive model

Selected Data Parameters	Unit
Rotations per minute	rev/min
Lithology	-
Gel 10min/Gel 10s	-
Pump pressure	psi
Hole size	in
Fracture pressure	psi
Pore pressure	psi
Northing	m
Easting	m
Fann 600/Fann 300	-

228

229 3.2 Mud Loss Prediction Using SVM

230 The predicted SVM models were built in MATLAB and compared to determine the best SVM model.
 231 The model with the lowest RMSE and highest R^2 score is regarded well performing. A R^2 value of 1
 232 indicates that the LC rate prediction is error-free. The Fine Gaussian SVM model produced the greatest R^2
 233 value of 0.91 for testing, indicating that it is the best SVM model for predicting LC rates in the Marun
 234 oilfield. This finding is consistent with the findings of Abbas et al. (2019), who used the same Gaussian
 235 SVM model selection. On the other hand, the Cubic SVM model is the second best SVM model for testing,
 236 with an R^2 value of 0.79. The Linear SVM model has the lowest performance score, with an R^2 value of
 237 0.23 for testing. The model's RMSE value was also greater when compared to other SVM models used.
 238 When compared to Abbas et al. (2019), the SVM model has a lower accuracy (R^2) value with an R^2 of 0.97
 239 for the training dataset and an R^2 of 0.95 for the testing dataset. This could be due to discrepancies in data
 240 processing and screening methodology. The data was pre-processed by using a transfer function to
 241 standardise the data and the *fscaret* package to justify the rank of the parameters. The discrepancy in
 242 accuracy could also be due to a variation in input parameters. Fig. A1 depicts the predicted vs true response
 243 of each SVM model, while Table 4 displays the RMSE and R^2 for all SVM.

244

Table 4: SVM model results

Model type	Train		Test	
	RMSE	R^2	RMSE	R^2
Linear SVM	86.17	0.23	86.09	0.23
Quadratic SVM	73.52	0.44	73.05	0.45
Cubic SVM	47.7	0.77	45.38	0.79
Fine Gaussian SVM	40.58	0.83	29.23	0.91
Medium Gaussian SVM	62.85	0.59	60.71	0.62
Coarse Gaussian SVM	85.27	0.25	84.99	0.25

245 3.3 Mud Loss Prediction Using ANN

246 The predictive ANN models were built in MATLAB and compared to determine the best ANN model.
 247 The model with the best performance must have the lowest RMSE and the highest R^2 . A R^2 value of 1
 248 indicates that the LC rate prediction is error-free. The Wide NN model got the greatest R^2 value of 0.99 for
 249 testing, indicating that it is the best ANN model for predicting LC rates in the Marun oilfield. The Medium
 250 NN model is the second-best ANN model for testing, with an R^2 value of 0.91. The Narrow NN model had
 251 the lowest performance score, with an R^2 value of 0.84 for testing. The R^2 value is comparable to the current
 252 Wide NN results when compared to a study by Behnoud far & Hosseini (2017). This demonstrates that the
 253 results agreed with the previous study. The predicted versus true response of the ANN models is depicted
 254 in Fig. A2.

255 **Table 5:** ANN model results

Model type	Train		Test	
	RMSE	R^2	RMSE	R^2
Narrow NN	42.57	0.81	39.3	0.84
Medium NN	34.8	0.87	28.96	0.91
Wide NN	21.56	0.95	11.29	0.99
Bilayered NN	36.6	0.86	31.83	0.89
Trilayered NN	45.24	0.79	31.72	0.90

256 3.4 Comparison Between SVM and ANN

258 When the best SVM and ANN models were evaluated, it was determined that the ANN model, Wide
 259 NN, outperformed the SVM model in terms of performance and accuracy. The findings contrast from those
 260 of Sabah et al. (2020), who similarly predicted LC rates in the Marun oilfield. The study employs MLP and
 261 LSSVM models, which are also components of ANN and SVM, respectively. They also used the Savitzky-
 262 Golay (SG) filter to reduce noise, whereas the current study used the interquartile range to exclude any
 263 outliers. In addition, findings from Abbas et al. (2019)'s study revealed that SVM models had a higher R^2
 264 value than the ANN model. This contradicts the current study as the higher accuracy model is an ANN
 265 model. It may be the result of using different data screening and processing methods. They used a genetic
 266 algorithm to rank the correlation strength of the parameters while this study uses Pearson's correlation
 267 coefficient. Nevertheless, the ANN model utilised in this work produced the highest accuracy ($R^2=0.99$)
 268 when compared to their SVM model ($R^2=0.95$) for testing data. It is demonstrated that the data processing
 269 and screening strategy utilised in the current study enables the ANN model to predict with more accuracy.
 270 As a result, the Wide NN model is chosen as the best model for predicting LC rates in oil and gas wells.

271 3.5 Comparison of Other ML algorithms

272 Agin et al., (2020) employed data mining as its intermediary ML method in another investigation. Data
273 mining works by identifying a function to model the data. In contrast to this study, which splits the data
274 into training and testing, the entire dataset was used to validate the model. According to Agin et al., (2020)
275 the accuracy of data mining for prediction is comparably worse, with an R2 value of 0.4029. As a result, it
276 can be stated that the Wide NN model used in this work is far superior for predicting LC rates in oil and
277 gas wells.

278 4.0 Conclusion

279 SVM and ANN models were built in this paper to forecast lost circulation rates in Marun oilfield wells.
280 The prediction model included 2820 data sets from 19 data groups obtained from daily drilling reports,
281 daily mud reports, and final well reports. Outliers were removed from the dataset before creating the
282 prediction model by estimating the interquartile range. The parameters were then graded using Pearson's
283 correlation coefficient to remove redundancy. This helps the ML models to be more accurate while
284 minimising noise and error. Lithology, RPM, Gel 10min/Gel 10s, pore pressure, fracture pressure, pump
285 pressure, hole size, northing, easting, and Fann 600/Fann 300 were chosen as parameters with strong
286 connection. Based on the results of the model evaluation, the following conclusions can be drawn:

- 287 • Wide NN outperformed other ANN models in terms of predicting lost circulation, with an RMSE
288 of 11.29. This was followed by Medium NN, which had an RMSE of 28.96.
- 289 • With an RMSE of 29.23, Fine Gaussian SVM was the best-performing SVM model.
- 290 • Wide NN from ANN is the best ML algorithm between ANN and SV.
- 291 • When the data has been processed to remove outliers and reduce noise in the data, the prediction
292 model can offer improved accuracy.
- 293 • Pearson's correlation coefficient ranking approach for parameters is effective for identifying
294 parameters that do not contribute positively to prediction model accuracy.

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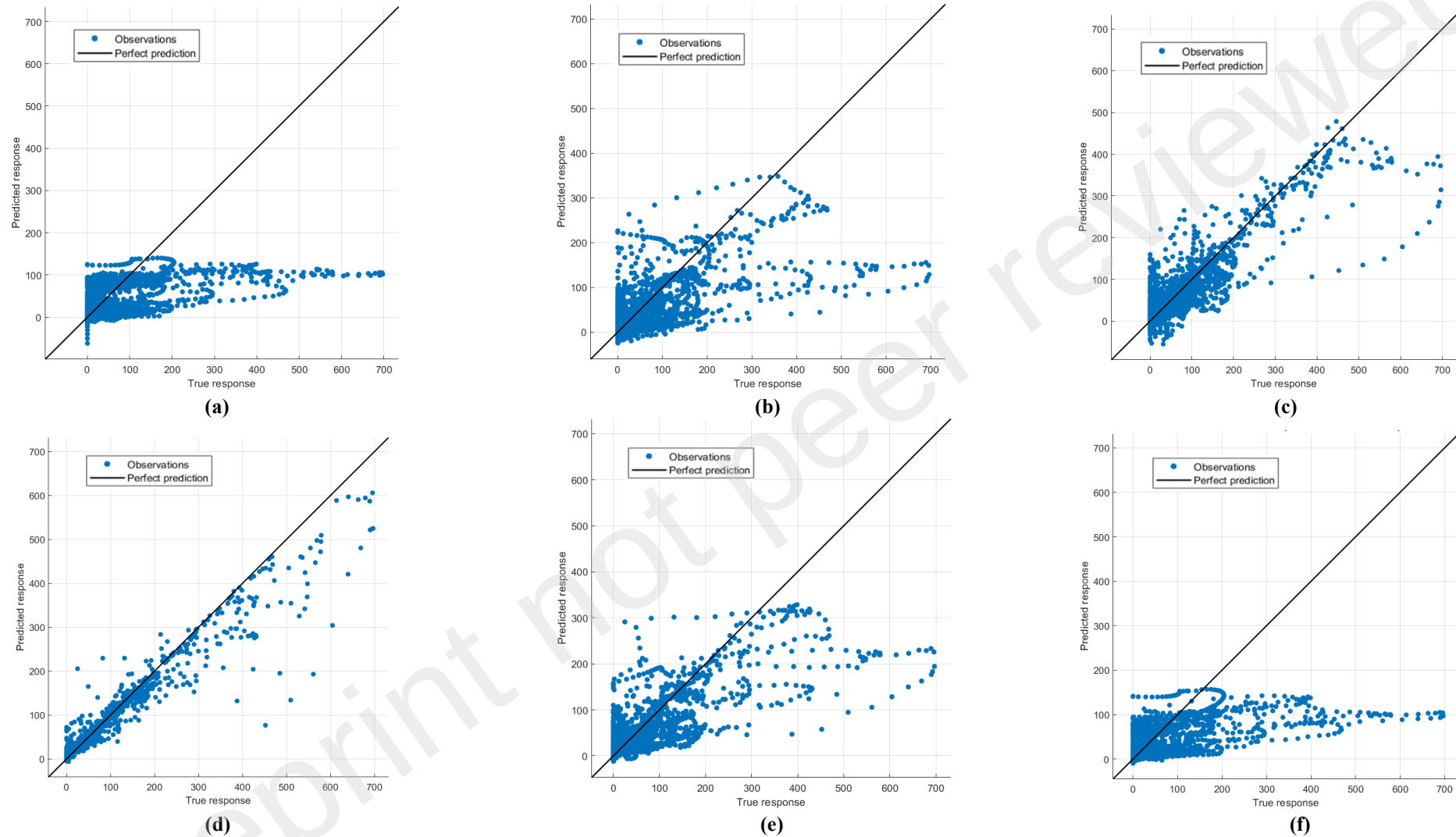


Fig A1: Predicted response vs true response for mud loss prediction for six SVM models applied to the testing data subsets: (a) Linear SVM; (b) Quadratic SVM; (c) Cubic SVM; (d) Fine Gaussian SVM; (e) Medium Gaussian SVM; (f) Coarse Gaussian SVM.

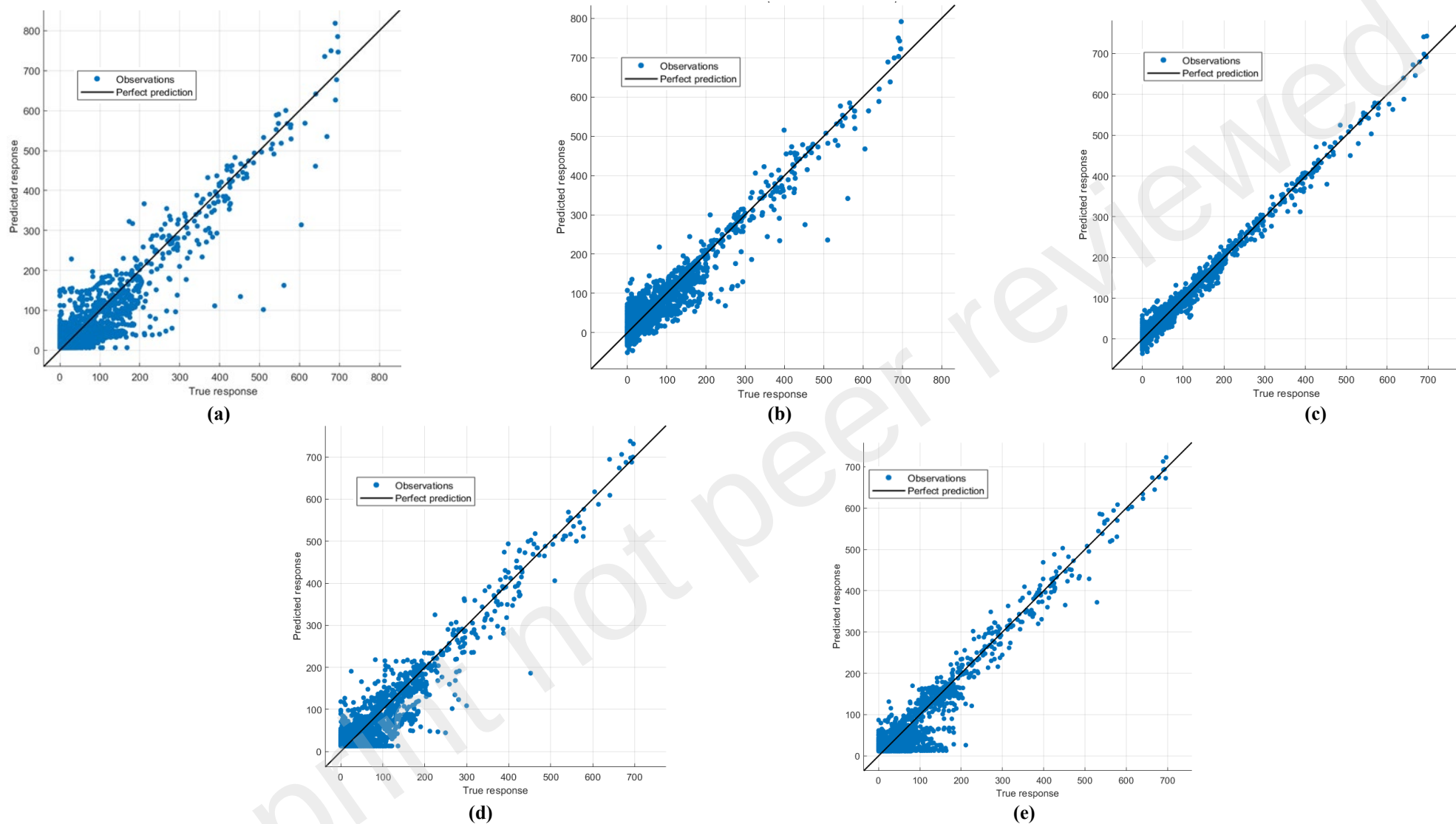


Fig A2: Predicted response versus true response for mud loss prediction for five ANN models applied to the testing data subsets:

(a) Narrow NN; (b) Medium NN; (c) Wide NN; (d) Bilayered NN; (e) Trilayered NN.